Manikya Rathna

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Education

University of Pittsburgh Aug 2024 – Apr 2026
Master of Science in Intelligent Systems (Artificial Intelligence) - GPA: 3.917/4.0

Visvesvaraya Technological University Aug 2017 – Aug 2021
Bachelor of Engineering in Electronics and Communication - GPA: 8.23/10.0

Technical Skills

Languages : Python, Java, Typescript, Javascript, HTML/CSS, MySQL, NoSQL, R Studio,React

LLM tools & NLP Tools: Hugging Face, QLora, Fine-tuning, Parameter Efficient Fine Tuning(PEFT), DPO, N-gram techniques

Deep Learning Frameworks : TensorFlow, Keras, PyTorch, BERT, Azure OpenAI, Langchain

Work Experience

AI Engineer Intern, Northbridge – Pittsburgh, USA (Link) Nov 2025 – Present

- Enhanced job-candidate matching accuracy by 35% as measured by application conversion rates, by developing a hybrid recommendation engine using AWS SageMaker, collaborative filtering, and NLP-based feature extraction from resumes and job descriptions.
- Automated visa-sponsorship classification for 6,000+ job postings with 73% accuracy, by fine-tuning BERT models on AWS EC2 and deploying serverless inference endpoints with Lambda and EFS for real-time employer profile analysis.
- Improved resume-to-job matching relevance by 28% as measured by student click-through rates, by deploying a transformer-based skill extraction model on AWS SageMaker and implementing approximate nearest neighbor search with OpenSearch for 4,500+ student profiles.

NLP Software Developer, University of Pittsburgh – Pittsburgh, USA Sept 2025 – Present

- Accomplished extraction of clinical insights from 135M+ patient notes with 92% accuracy, as measured by successful identification of 19,000+ tendinopathy cases (11,500 Achilles, 7,500 Patellar), by doing development of automated NLP pipelines using OHNLP and Python scripts.
- Streamlined ETL pipelines integrating 6 EHR clinical domains from 16+ CSV files, achieving 65% reduction in processing time, using Python multiprocessing, PostgreSQL COPY, and batch insert strategies.
- Optimized query performance on 135M+ row clinical notes database infrastructure, as measured by 70% improvement in query execution time and 40% reduction in data retrieval latency, by designing PostgreSQL indexing strategies (B-tree, GIN), table partitioning, and ICD-code filtering logic for targeted tendinopathy extraction.

Fellowship, Premium Automation and Premium Labs – Pittsburgh, USA (Link) Mar 2025 – May 2025

- Architected an end-to-end AI pipeline integrating LLMs (GPT, Gemini, Perplexity) with a Retrieval-Augmented Generation (RAG) framework to automate industrial market analysis, achieving a 70% reduction in manual research effort.
- Engineered a scoring mechanism to quantitatively evaluate stakeholder responses and market signals, visualizing insights via dynamic graphs to support executive-level decision-making.
- Architected a RAG-based NLP framework leveraging static and contextual embeddings to identify semantically relevant keywords across diverse process types and industries.

Graduate Research Assistance, University of Pittsburgh – Pittsburgh, USA Feb 2025 – May 2025

- Performed data analysis to identify trends and correlations in chewing dynamics as part of the DIETS research.
- Preprocessed sensor data and manually annotated chewing vs. non-chewing segments and focused on time-series data modeling, real-time signal processing, and deep learning evaluation.
- Conducted experiments using the WT901 multi-sensor to detect variations in chewing under Dr. Pengfei Zhou.

Machine Learning Engineer, Tata Consultancy Services – Bangalore, India Aug 2021 – Jul 2024

- Developed predictive models using XGBoost and Random Forest to accurately forecast home insurance premiums, enabling data-driven underwriting decisions and improving operational efficiency.
- Managed the design and execution of scalable ETL pipelines in AWS Glue, leveraging PySpark for large-scale data transformation, ensuring high-quality input for downstream ML models.
- Took the lead in integrating Spark with NoSQL databases to handle and analyze large, unstructured datasets, supporting complex queries and entity-level insights.
- Designed a suite of Flask based RESTful APIs to seamlessly deliver ML outputs to front-end applications, improving accessibility and response times.

AI Intern, Tequed Labs – Bangalore, India Mar 2021 – Apr 2021

- Built a chatbot capable of interacting with a SQL dataset containing over 300k rows, utilizing LangChain tools.
- Devised innovative feature extraction techniques that enhanced the quality of insights derived from large datasets; methodologies implemented across three critical projects yielded improved customer satisfaction scores.

Projects

Biomedical Knowledge Graph with Proof-Path Similarity Discovery Aug 2025 - Dec 2025

- Developed platform constructs a unified biomedical knowledge graph by integrating multiple ontologies with automated cross-reference extraction from structured fields, free-text definitions, and synonym annotations.It employs Proof-Path Similarity Discovery methodology to identify non-obvious relationships through configurable graph traversal algorithms.

Uncertainty-Aware Preference Distillation with Test-time Reward Ensemble Jan 2025 - Apr 2025

- Developed and benchmarked a framework for LLM alignment, leveraging test-time reward ensembles and epistemic uncertainty quantification to enhance robustness in RLHF, achieving superior performance on AlpacaEval 2.0 with Llama3.2-3B and Qwen2.5-7B models.

Intelligent Device Control Using Brain Machine Interface Sept 2022 - Dec 2022

- Developed a ML model for EEG-based brain-actuated wheelchair with the International 10-20 electrode setup, enabling directional control, cursor and limb movement, prosthesis control, and communication via a Brain-Computer Interface (BCI). Published at Proceedings of ERCICA 2023 (Vol. 1) Springer Nature Singapore Pte Ltd, 2023. (Link)

Awards

- Awarded for leading a UI development project that enhanced user engagement, resulting in a measurable increase in daily active users and improved user retention at Tata Consultancy Services.
- Awarded with KSCST (Karnataka State Council for Science and Technology) Student Project Award for project "BCI Model to assist Aged or Physically Challenged People", with additional funding from the KSCST trust.